L.D. COLLEGE OF ENGINEERING, AHMEDABAD

Department of Instrumentation & Control Engineering

MATLAB/Simulink Simulation Project Report

Performance Comparison of Conventional PID and Fuzzy-PID Control for a Nonlinear Liquid-Level System

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Abstract

Liquid-level control is a fundamental requirement in process industries such as chemical processing, water treatment, and petroleum refining, where accurate regulation of tank levels directly affects product quality, safety, and downstream process stability. Many liquid-level processes exhibit nonlinear dynamics, particularly when outflow is governed by gravity-driven discharge through an orifice, and such nonlinearity makes it difficult for a fixed-gain conventional controller to maintain uniformly good performance across the full operating range.

This project investigates the performance of a conventional Proportional-Integral-Derivative (PID) controller and a hybrid fuzzy-PID controller applied to a nonlinear liquid-level process. The nonlinear tank dynamics, in which the outflow rate is proportional to the square root of the liquid level, were modelled in MATLAB/Simulink. A Mamdani-type fuzzy inference system was designed to generate a supplementary correction signal based on the instantaneous error and the rate of change of error, and this correction was added to the output of the conventional PID controller to form the hybrid fuzzy-PID structure.

Both controllers were evaluated for a step change in the level setpoint. The conventional controller produced a maximum peak level of approximately 2.9905 against a setpoint of 2.7, corresponding to a maximum overshoot of approximately $10.76\,\%$. The fuzzy-PID controller reduced the maximum peak to approximately 2.8199, corresponding to a maximum overshoot of approximately $4.44\,\%$, while both controllers settled to essentially zero steady-state error. For the specific simulation conditions tested, the fuzzy-PID controller therefore reduced the maximum overshoot by approximately $59\,\%$ relative to the conventional controller. This result is reported strictly for the tested case and is not presented as a general claim of superiority of fuzzy-PID control over conventional PID control.

1. Introduction

Liquid-level control is one of the most common control problems encountered in process industries. Storage tanks, reactors, boilers, and separation columns all require the liquid level to be maintained at, or driven to, a desired setpoint in order to ensure safe and efficient plant operation. Deviations from the desired level can lead to overflow, insufficient residence time, or disruption of downstream processes, making accurate level control an important practical objective.

Feedback control is the standard approach used to regulate liquid level. In a feedback control loop, the measured level is compared with the desired setpoint to generate an error signal, and the controller acts on this error to adjust the manipulated inflow so that the error is driven towards zero. The Proportional-Integral-Derivative (PID) controller is the most widely used feedback controller in industrial practice because of its simple structure, well-understood tuning methods, and generally acceptable performance for a large class of processes.

However, a conventional PID controller uses fixed gains that are typically tuned around a single operating point. Many real processes, including gravity-drained tanks, are nonlinear because the outflow rate depends on the square root of the liquid head. As the operating level changes, the effective process gain changes as well, and a PID controller tuned for one operating region may produce excessive overshoot or sluggish response in another. This limitation motivates the use of supplementary control strategies that can adapt to nonlinear process behavior without discarding the reliability of the PID structure.

Fuzzy logic control offers one such supplementary approach. A fuzzy controller uses linguistic rules, expressed in terms of qualitative variables such as "Negative", "Zero", and "Positive", to compute a control action from the error and the change in error. Because the fuzzy rule base can express nonlinear input-output relationships in a straightforward and interpretable way, fuzzy logic is well suited to providing a corrective action that compensates for the nonlinearity of the process without requiring an explicit mathematical model of that nonlinearity.

In this project, a fuzzy correction signal is combined with the output of a conventional PID controller to form a hybrid fuzzy-PID controller. The conventional PID controller retains responsibility for the baseline regulation and for eliminating steady-state error through integral action, while the fuzzy logic block supplies an additional correction that responds to the instantaneous error and its rate of change. A nonlinear liquid-level tank model, in which the outflow is proportional to the square root of the level, is used as the test process because it captures the essential nonlinearity found in many gravity-drained industrial tanks and therefore provides a realistic basis for comparing the two control strategies.

2. Project Objectives

1. To develop a nonlinear liquid-level process model in MATLAB/Simulink.
2. To design and simulate a conventional PID controller.
3. To design a Mamdani fuzzy logic controller.
4. To integrate fuzzy logic with PID to form a hybrid fuzzy-PID controller.
5. To evaluate transient response for a step change in liquid-level setpoint.
6. To compare maximum overshoot and steady-state behavior of conventional PID and fuzzy-PID.
7. To investigate whether fuzzy correction can improve the transient response of the nonlinear process.

3. System Model – Nonlinear Liquid-Level Process

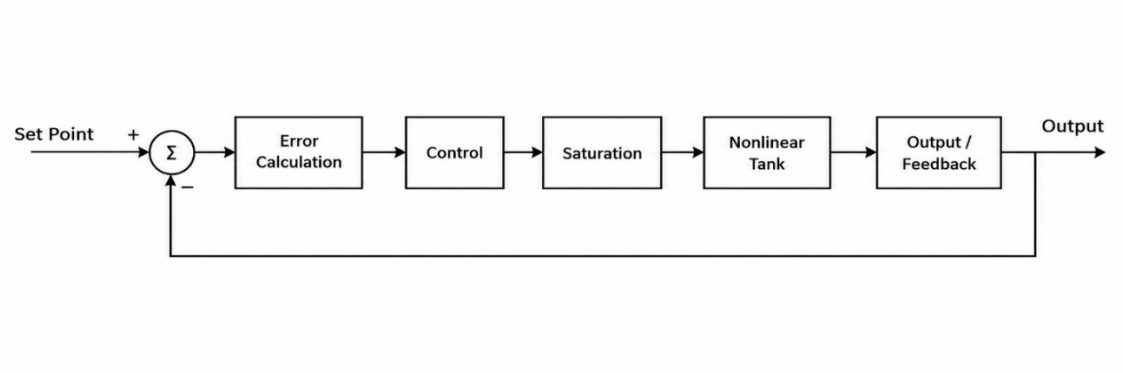
The nonlinear liquid-level process considered in this project is governed by the first-order nonlinear differential equation

$$\frac{dh}{dt} = u - \sqrt{h}$$

where h is the liquid level in the tank and u is the manipulated inflow (control input). The outflow term is proportional to the square root of the liquid head, consistent with gravity-driven discharge through a fixed orifice, and it is this square-root relationship that introduces nonlinearity into the process: the effective process gain varies with the operating level, so a controller tuned at one level will not necessarily behave identically at another.

3.1 Simulink Implementation

The overall signal flow implemented in Simulink is as follows:



The nonlinear tank subsystem was built using the following blocks:

- A summing junction to compute the error between the setpoint and the measured level.
- A PID controller block acting on the error signal.
- A saturation block limiting the controller output to a physically realistic range.
- A second summing junction implementing $u - \sqrt{h}$.
- An integrator block $(1/s)$ that integrates the net flow to obtain the liquid level h.
- A square-root block that computes the nonlinear outflow term from the current level.
- A feedback path returning the liquid level to the error-calculation junction.

An additional saturation block was placed immediately before the square-root block. This was necessary because, during transient conditions, the signal entering the square-root block could momentarily become negative; since the square root of a negative number is not defined for a real-valued simulation, this saturation block constrains the input to the square-root function to non-negative values and prevents the simulation from producing invalid results.

3.2 Simulation Parameters

Parameter	Value
Step initial value	0
Step final value	2.7
Step time	1 s
Simulation stop time	60 s
Controller output saturation	0 to 2
Liquid-level feedback saturation	0 to 10

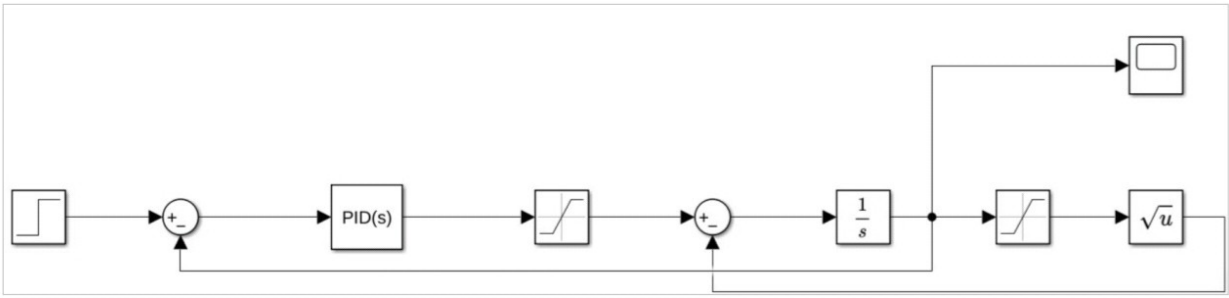


Figure 1 : Simulink model of the nonlinear liquid-level control system.

4. Conventional PID Controller (PI Configuration)

The conventional controller used in this project is the standard Proportional-Integral-Derivative (PID) controller, whose time-domain control law is given by

$$u(t) = K_p e(t) + K_i \int e(t)\,dt + K_d \frac{de(t)}{dt}$$

where $e(t)$ is the instantaneous error between the setpoint and the measured liquid level, and K_p, K_i, and K_d are the proportional, integral, and derivative gains respectively.

For this project, the conventional controller parameters were set as follows:

Gain	Value
K_p	1
K_i	0.5
K_d	0

Since the derivative gain K_d was set to zero, this implementation effectively behaves as a Proportional-Integral (PI) controller, although it was retained within the standard PID controller block structure in Simulink for consistency with the fuzzy-PID implementation described later. The integral term is essential for this application because it accumulates the error over time and continues to act as long as any error persists; this drives the steady-state error to zero even in the presence of the process nonlinearity, whereas a purely proportional controller would leave a residual offset between the setpoint and the final level.

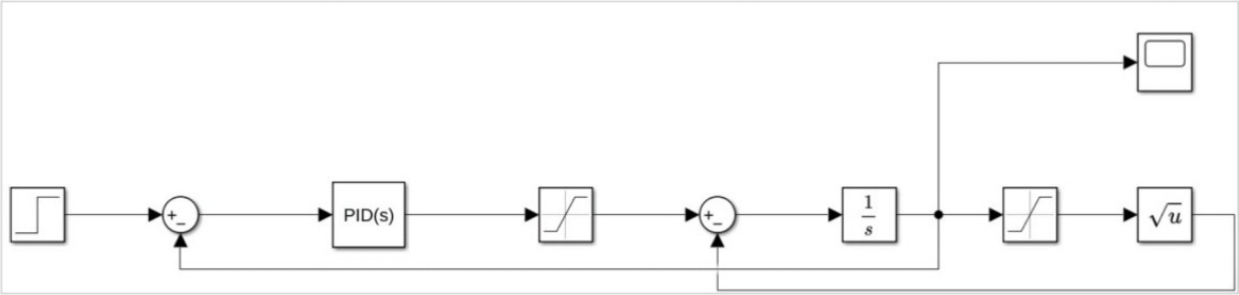
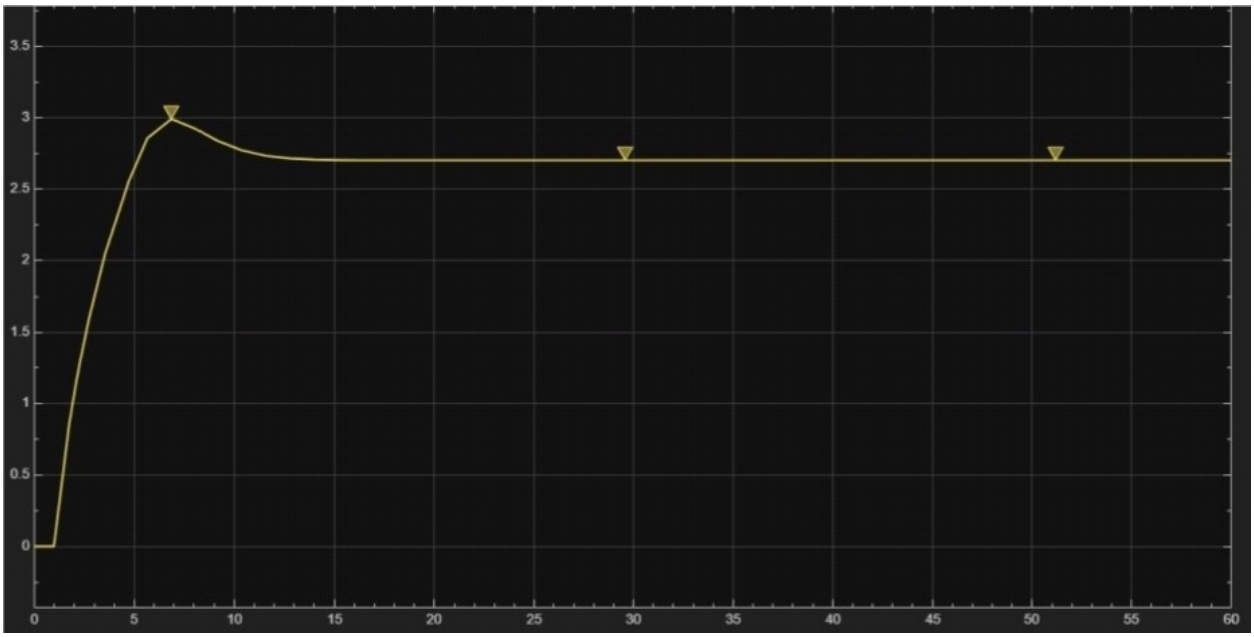


Figure 2. Conventional PID controller implementation in MATLAB/Simulink.



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Figure 3. Step response of the nonlinear liquid-level system using the conventional controller.

4.1 Conventional PID – Observed Results

With the setpoint set to 2.7, the following values were read from the Scope trace:

Quantity	Value
First peak time	~ 6.1754 s
First peak value	~ 2.9905
Later observed peak/value	~ 2.8000
Final value	~ 2.7

The maximum percentage overshoot is calculated using

$$\%\ \text{Overshoot} = \left[\frac{\text{Maximum Peak} - \text{Setpoint}}{\text{Setpoint}}\right] \times 100$$

Substituting the maximum peak value of 2.9905 and the setpoint of 2.7:

$$\%\ \text{Overshoot} = \left[\frac{2.9905 - 2.7}{2.7}\right] \times 100 \approx 10.76\,\%$$

The response in Figure 3 rises quickly from zero, overshoots the setpoint, and then settles into the vicinity of the setpoint over the remainder of the 60-second simulation window. Because the exact time at which the response enters and remains within a narrow tolerance band cannot be reliably read from the scope screenshot, an exact settling time is not reported; the response can be described qualitatively as settling to approximately 2.7 within the simulated 60-second window.

5. Fuzzy Logic Controller Design

A Mamdani Type-1 fuzzy inference system (FIS) was designed to generate a correction signal that supplements the conventional PID controller. The fuzzy controller has two inputs and one output:

- Input 1: Error - the difference between the setpoint and the measured liquid level.
- Input 2: Change in Error - the rate at which the error is varying.
- Output: Δu - the fuzzy correction added to the PID output.

Each input and the output are represented using three triangular membership functions, labelled Negative, Zero, and Positive, defined over the normalized universe of discourse $[-1, 1]$:

Membership Function	Triangular Parameters [a b c]
Negative	[-1 -1 0]
Zero	[-1 0 1]
Positive	[0 1 1]

A negative error indicates that the measured level exceeds the setpoint, a zero error indicates that the level is at (or very near) the setpoint, and a positive error indicates that the level is below the setpoint. Similarly, a negative change in error indicates that the error is decreasing (the response is moving towards the setpoint), a zero change in error indicates that the error is roughly constant, and a positive change in error indicates that the error is increasing (the response is moving away from the setpoint). The fuzzy output Δu represents a corrective adjustment to the control signal: a negative Δu reduces the control effort, a positive Δu increases it, and a zero Δu leaves the PID output largely unmodified.

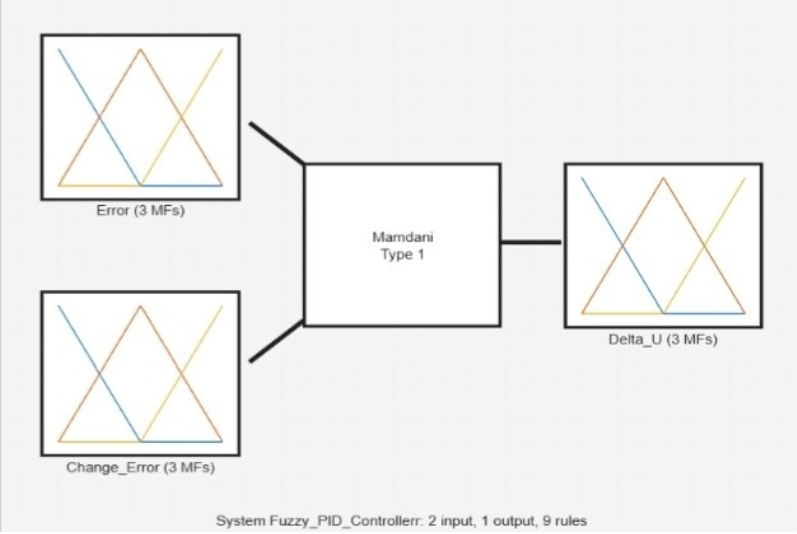


Figure 4. Mamdani fuzzy inference system designed for the controller.

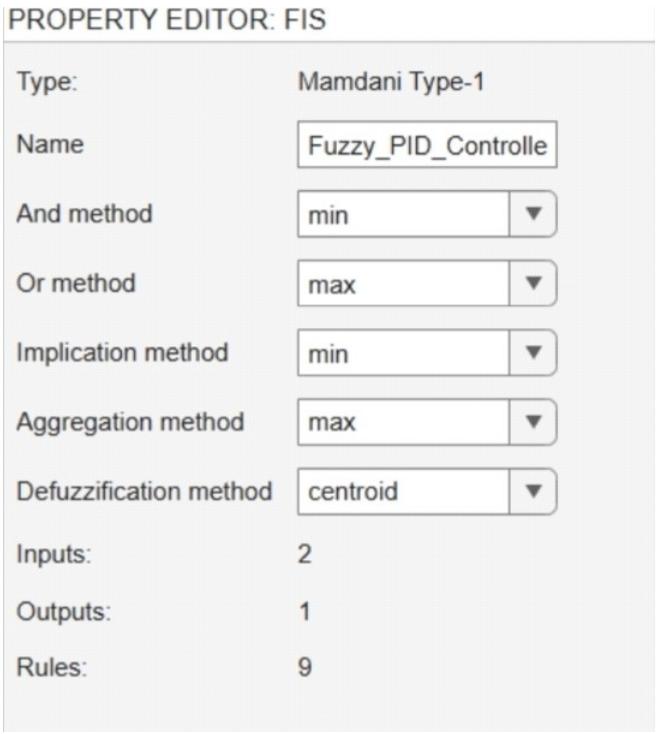


Figure 5. Fuzzy inference system property editor showing Mamdani Type-1 configuration (min/max/centroid methods, 2 inputs, 1 output, 9 rules).

5.1 Rule Base

The fuzzy controller uses a complete rule base of nine rules covering every combination of the three membership functions for Error and Change in Error. The rule base is summarized in Table 5.

Error	Change in Error	Delta_U
Negative	Negative	Negative
Zero	Negative	Negative
Positive	Negative	Zero
Negative	Zero	Negative
Zero	Zero	Zero
Positive	Zero	Positive
Negative	Positive	Zero
Zero	Positive	Positive
Positive	Positive	Positive

	Rule	Weight	Name
1	If Error is Negative and Change_Error is Negative then Delta_U is Negative	1	rule1
2	If Error is Zero and Change_Error is Negative then Delta_U is Negative	1	rule2
3	If Error is Positive and Change_Error is Negative then Delta_U is Zero	1	rule3
4	If Error is Negative and Change_Error is Zero then Delta_U is Negative	1	rule4
5	If Error is Zero and Change_Error is Zero then Delta_U is Zero	1	rule5
6	If Error is Positive and Change_Error is Zero then Delta_U is Positive	1	rule6
7	If Error is Negative and Change_Error is Positive then Delta_U is Zero	1	rule7
8	If Error is Zero and Change_Error is Positive then Delta_U is Positive	1	rule8
9	If Error is Positive and Change_Error is Positive then Delta_U is Positive	1	rule9

Figure 6. Rule base of the fuzzy controller, as configured in the Rule Editor.

This rule base provides a nonlinear correction that depends jointly on the magnitude and direction of the control error and on whether the error is growing or shrinking. For example, when the error is positive and increasing (Positive, Positive), the rule base calls for a positive correction to accelerate the approach to the setpoint; when the error is negative and decreasing (Negative, Negative), the rule base calls for a negative correction to help reduce overshoot as the response returns towards the setpoint.

6. Fuzzy-PID Hybrid Controller

The final controller implemented in this project is not a complete replacement of the PID controller. Instead, it is a hybrid structure in which the fuzzy inference system supplies a supplementary correction that is added to the output of the conventional PID controller:

$$u = u_{PID} + \Delta u_{fuzzy}$$

where u_{PID} is the output of the conventional PID controller and Δu_{fuzzy} is the fuzzy correction term. The combined signal is passed through the same actuator saturation block used in the conventional controller before being applied to the nonlinear tank process.

The PID gains used within the hybrid controller were kept identical to those used in the conventional controller, so that any difference in performance can be attributed to the fuzzy correction rather than to re-tuning of the PID gains:

Gain	Value
K_p	1
K_i	0.5
K_d	0

Two scaling gains were used to condition the signals entering and leaving the fuzzy inference system:

- Error scaling gain (applied to the error before it enters the fuzzy controller): $1/3$.
- Fuzzy output scaling gain (applied to Δu before it is added to the PID output): 0.5.

The error signal is used, after scaling, as the first fuzzy input, while the change in error (computed from the same error signal) is used as the second fuzzy input. The signal flow through the hybrid controller can be summarized as follows:

- Error -> scaling $(1/3)$ -> fuzzy input 1.
- Change in Error -> fuzzy input 2.
- Fuzzy inference (Mamdani, 9 rules) -> Δu.
- Δu -> output scaling gain (0.05).
- PID output + scaled fuzzy correction -> actuator saturation -> nonlinear liquid-level process.

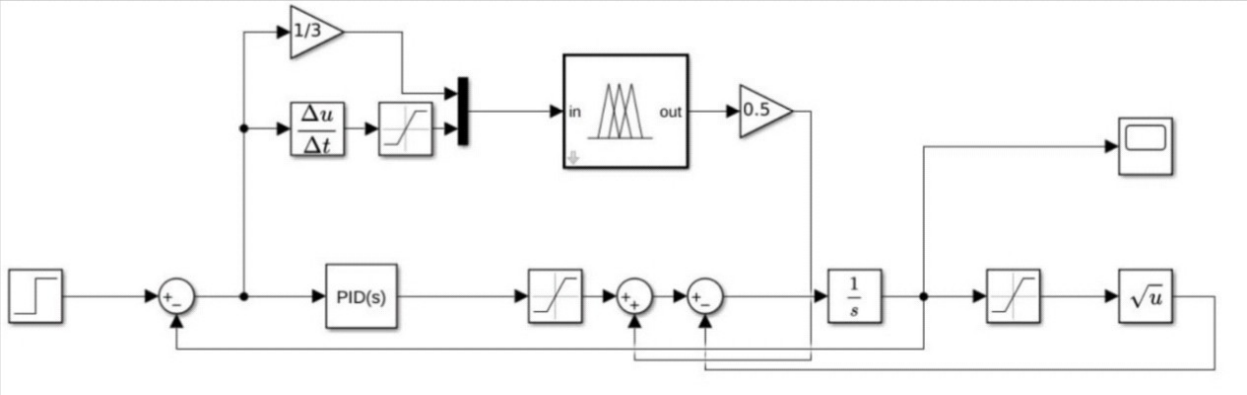


Figure 7. Hybrid fuzzy-PID controller implementation in MATLAB/Simulink.

7. Fuzzy-PID Simulation Result

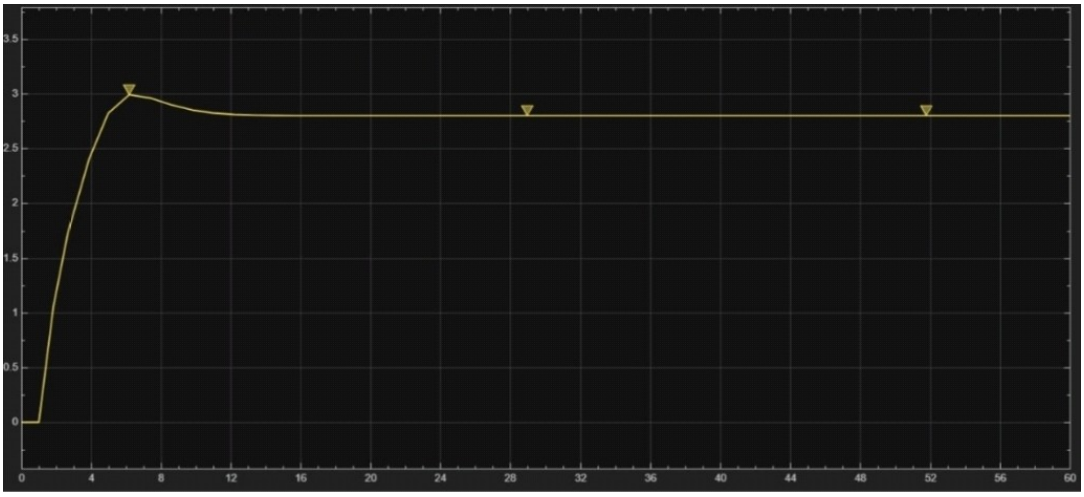


Figure 8. Step response of the nonlinear liquid-level system using the fuzzy-PID controller.

With the setpoint set to 2.7, the following values were read from the Scope trace of the fuzzy-PID simulation:

Quantity	Value
First peak time	~ 6.69 s
First peak value	2.8199
Later observed value	2.7000
Final value	2.7000

The maximum percentage overshoot for the fuzzy-PID controller is calculated as

$$\%\ \text{Overshoot} = \left[\frac{2.8199 - 2.7}{2.7}\right] \times 100 \approx 4.44\,\%$$

As shown in Figure 8, the fuzzy-PID response rises to a lower peak than the conventional response, then returns to and remains at the setpoint of 2.7, indicating essentially zero steady-state error. As with the conventional case, an exact settling time is not claimed from the available screenshot; the response is described qualitatively as returning to the vicinity of the setpoint within the 60-second simulation window.

8. Performance Comparison

Parameter	Conventional PID	Fuzzy-PID
K_p	1	1
K_i	0.5	0.5
K_d	0	0
Setpoint	2.7	2.7
Maximum peak	2.9905	2.8199
Approx. maximum overshoot	10.76%	4.44%
First peak time	6.1754 s	~ 6.69 s
Final value	~ 2.7	2.7000
Steady-state error	~ 0	~ 0

The percentage reduction in maximum overshoot achieved by the fuzzy-PID controller relative to the conventional controller is calculated as :

$$\text{Overshoot reduction} = \left[\frac{10.76 - 4.44}{10.76}\right] \times 100 \approx 58.7\,\%$$

For the tested simulation, the fuzzy-PID controller therefore reduced the maximum overshoot by approximately $59\,\%$ relative to the conventional PID controller. It should be noted that the available results show the conventional response reaching the final region of its response earlier in time than the fuzzy-PID response; the main demonstrated advantage of the fuzzy-PID controller in this project is therefore a reduction in maximum overshoot and an improvement in transient behavior, and not an improvement in settling time. Both controllers achieve essentially zero steady-state error.

9. Results and Discussion

The simulation results obtained in this project allow a direct, like-for-like comparison of the conventional PID controller and the hybrid fuzzy-PID controller under identical process conditions, identical PID gains, and an identical step change in setpoint. The following observations summarize the key findings:

1. Both controllers successfully track the desired liquid-level setpoint of 2.7.
2. Integral action, present in both controller structures, results in essentially zero steady-state error for both cases.
3. The conventional controller produces a maximum peak of approximately 2.9905, corresponding to approximately 10.76% overshoot.
4. The fuzzy-PID controller reduces the maximum peak to approximately 2.8199.
5. The fuzzy-PID controller reduces the maximum overshoot to approximately 4.44%.
6. This represents an approximate 59% reduction in maximum overshoot relative to the conventional controller.
7. The fuzzy correction modifies the transient response according to the instantaneous error and its rate of change, rather than relying solely on the fixed proportional and integral action of the PID controller.
8. The fuzzy-PID response still converges to the desired final value of 2.7.
9. The fuzzy controller does not necessarily improve every performance metric; in this particular simulation, the principal demonstrated improvement is the reduction in maximum overshoot.
10. Further tuning of the membership functions, rule base, scaling factors, and PID gains could potentially yield improvements in settling time and other performance metrics.

Overall, the results indicate that adding a fuzzy correction term to a conventional PID controller can meaningfully reduce transient overshoot in a nonlinear liquid-level process, without compromising the steady-state accuracy provided by the integral action of the PID controller. This improvement is specific to the tested simulation conditions — a single step change from 0 to 2.7 applied to the particular nonlinear tank model, PID gains, membership functions, rule base, and scaling factors used in this project — and should not be interpreted as a general claim that fuzzy-PID control is superior to conventional PID control under all conditions.

10. Advantages of the Proposed Fuzzy-PID Approach

- Reduced transient overshoot in the tested case.
- Better handling of the process nonlinearity through rule-based correction rather than a single fixed gain.
- The rule base provides a nonlinear correction that can be tailored to the direction and magnitude of the error.
- The reliable integral action of the underlying PID controller is retained, preserving zero steady-state error.
- The approach does not require a complete replacement of the conventional PID structure already familiar to plant operators.
- The fuzzy correction can be further tuned for different operating conditions without redesigning the entire controller.

11. Limitations

- This work is a simulation study only; no physical tank prototype was implemented or tested.
- The results depend on the selected membership functions and fuzzy rules, which were chosen based on engineering judgement rather than a formal optimization procedure.
- The scaling factors applied to the fuzzy inputs and output influence the stability and performance of the hybrid controller and were not exhaustively tuned.
- Only one nonlinear process model and one step-setpoint condition (0 to 2.7) were tested.
- Further testing with external disturbances, parameter variations, and different setpoint magnitudes would be required to draw stronger conclusions.
- The fuzzy controller did not improve every performance metric in the current tuning; in particular, it did not demonstrate an improvement in settling time.

12. Future Scope

The following directions were not implemented in the present project but are identified as logical extensions of this work:

- Adaptive fuzzy-PID tuning, in which the scaling factors or PID gains are adjusted online.
- Optimization of the membership functions and rule base using systematic design methods.
- PID gain tuning using metaheuristic methods such as Genetic Algorithm or Particle Swarm Optimization.
- Testing the controller's ability to reject external disturbances.
- Evaluating performance across multiple operating points and setpoint magnitudes.
- Real-time implementation using a microcontroller, ESP32, or PLC platform.
- Hardware-in-the-loop testing prior to physical deployment.

- Construction of an experimental liquid-level tank setup for physical validation.
- Comparison with other advanced control strategies such as Model Predictive Control (MPC) or sliding-mode control.

13. Conclusion

In this project, a nonlinear liquid-level control system, with outflow proportional to the square root of the liquid level, was successfully modelled in MATLAB/Simulink. A conventional PID (effectively PI, with $K_d = 0$) controller and a hybrid fuzzy-PID controller, in which a Mamdani fuzzy inference system supplies a supplementary correction to the PID output, were both implemented and simulated for an identical step change in setpoint from 0 to 2.7.

Both controllers achieved approximately zero steady-state error at the 2.7 setpoint. The conventional controller produced a maximum peak of approximately 2.9905, corresponding to a maximum overshoot of approximately $10.76\,\%$, while the fuzzy-PID controller produced a maximum peak of approximately 2.8199, corresponding to a maximum overshoot of approximately $4.44\,\%$. The fuzzy-PID controller therefore reduced the maximum overshoot by approximately $59\,\%$ relative to the conventional controller in the tested simulation.

These results demonstrate that a fuzzy correction, added to a conventional PID controller, can improve the transient behavior of a nonlinear liquid-level system by reducing overshoot, while preserving the steady-state accuracy provided by the PID integral action. The finding is specific to the simulation conditions used in this project, and further tuning together with experimental validation on a physical tank would be required before broader conclusions could be drawn.

14. Key Findings

Metric	Value
Setpoint	2.7
Conventional PID (PI) – maximum peak	2.9905
Conventional PID (PI) – maximum overshoot	10.76%
Fuzzy-PID – maximum peak	2.8199
Fuzzy-PID – maximum overshoot	4.44%
Reduction in maximum overshoot	˜ 58.7%
Final value (both controllers)	˜ 2.7
Steady-state error (both controllers)	˜ 0

15. References

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